David Osorio

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Education

Wesleyan University — Middletown, CT
Bachelor of Arts in Computer Science and Mathematics (GPA: 3.71 / 4.0) — *May 2028*
- **Relevant Coursework:** Discrete Mathematics, Applied Data Analysis, Computer Science I/II

Houston City College — Houston, TX
Associate of Science in Computer Science (GPA: 3.97 / 4.0) — *May 2025*
- Graduated with Highest Honors | Honors College Graduate
- **Relevant Coursework:** Data Structures and Algorithms, Computer Organization and Architecture

Research Experience

NSF Undergraduate Researcher — Houston, TX
University of Houston-Downtown — *June 2026 – July 2026*
- Architected privacy-preserving Python data pipelines utilizing native NumPy and pandas to process complex UCI machine learning datasets, quantifying the tradeoff between Local Differential Privacy (LDP) mechanisms including Direct, Symmetric, and Optimal Unary Encoding and baseline Naive Bayes classification utility.
- Engineered custom dimensionality reduction (PCA/DCA) algorithms and equal-width discretization frameworks to benchmark statistical accuracy retention and data security across continuous clinical datasets under dynamic privacy budgets.

Student Researcher — Middletown, CT
Wesleyan University — *January 2026*
- Spearheaded quantitative climate change research on a national U.S. survey ($N = 929$), analyzing demographic and regional variables to identify key statistical predictors.
- Executed advanced bivariate and multivariate statistical analyses including Chi-square, ANOVA, and logistic/linear regressions using R to extract actionable data insights.

Projects

NYC Crime Explorer | *Python, pandas, SciPy, Frontend Web Maps*
- Developed an interactive data visualization web application merging NYPD criminal complaint records with U.S. Census API data to geographically map offense severity.
- Engineered dynamic frontend filters and a Python backend utilizing pandas and SciPy to execute real-time regression analyses controlling for key demographic variables.

Stock Prediction Agent | *Python, Yahoo Finance API, pandas, Matplotlib*
- Designed an automated quantitative trading script in Python utilizing the Yahoo Finance API and a comprehensive backtesting framework to simulate historical market data models and forecast equity price movements.
- Engineered a real-time analytical module for automated equity trading, generating actionable insights and optimizing risk management based on historical volatility metrics.

Work Experience

Service Desk Student Worker — Middletown, CT
Wesleyan University — *September 2025 – Present*
- Delivered Tier 1 IT support, independently resolving critical identity and access management incidents across Duo Security, Microsoft Outlook, and Google Workspace.
- Diagnosed and resolved diverse hardware and software malfunctions for students and faculty, ensuring continuous uptime for campus network systems and printing devices.

IT Technician — Houston, TX
University of Houston at Katy — *October 2024 – August 2025*
- Maintained and troubleshot complex hardware, software, and audio-visual systems across high-volume university lecture halls to facilitate seamless classroom instruction.
- Administered critical campus IT infrastructure, optimizing network connectivity and actively troubleshooting systems for library computing and printing networks.

Honors & Awards

Citation of Citizenship in the Department of Mathematics and Computer Science	May 2026
ASA DataFest CT 2026 Category Winner	Apr. 2026
Ronald E. McNair Scholar	Nov. 2025
Phi Theta Kappa Membership	Feb. 2025
Houston City College Honors College Entry/Graduation	July 2024
AP Scholar with Distinction	July 2024
Texas First Early High School Graduation Scholar	May 2024

Technical Skills & Additional

Technical Skills: R, RStudio, C, C++, Python, Windows Server, Network Management, Access Control Lists

Certifications & Training: CompTIA Network+ CE, Rice University Environmental Data Academy

Languages: English (Fluent), Spanish (Fluent)